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Photonic Integrated Circuits with Grating Couplers Emitting Low Divergence Beams on or adjacent to Movable Platforms for Augmented Reality Glasses and LiDAR

Abstract

Laser-scanning systems for augmented reality glasses displays and LiDAR (light detection and ranging) use photonic integrated circuits to generate multiple collimated (or low-divergence) modulated laser beams. Laser light is guided on the photonic chip, expanded, and emitted from the surface of the chip (using grating coupler devices) as one or more large-diameter (e.g., millimeter-scale), collimated or low-divergence beams. These gratings may be integrated onto movable microelectromechanical systems (MEMS) plates to steer the beam along two angular axes. Alternatively, the beam(s) from the grating(s) may be steered by a separate MEMS mirror integrated onto the photonic chip or on another chip. Such systems enable multi-beam laser scanning systems without collimation lenses, reducing the size and packaging complexity of high-performance laser scanning systems for augmented reality glasses and LiDAR.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application claims the priority benefit, under 35 U.S.C. 119(e), of U.S. Application No. 63/562,413, filed Mar. 7, 2024, which is incorporated herein by reference in its entirety for all purposes.

BACKGROUND

[0002] Head-mounted displays and near-eye displays for augmented reality (also referred to as mixed reality or extended reality) are transparent (see-through) glasses that overlay images generated by light engines onto the user's field of view (FOV). The light engines are typically on the periphery of the glasses and the light is guided and directed toward the user's eyes using optical combiners. An optical combiner is an optical device placed close to the eye(s) of a user, allowing the user to see augmented content, such as text or images, as well as the real world behind the optical combiner in the viewing direction of the user, at the same time. The size and weight of the optical components and systems are critical to head-mounted displays.

[0003] Laser-scanning light engines have become the focus of a growing number of commercial efforts for head-mounted displays. Laser-scanning has advantages of increased brightness, image contrast, and power efficiency compared to panel microdisplays.

[0004] The laser-scanning approach has multiple performance trade-offs and limitations, including:

[0005] 1. For fixed fast axis mirror frequency and maximum laser modulation rate, there is a trade-off between FOV and resolution, i.e., for each displayed frame, there is a maximum number of resolvable spots that can be defined due to the limited mirror and laser modulation speeds. [0006] 2. There are practical limitations on the fast axis mirror frequency. Increasing the fast axis mirror frequency requires a reduction in mirror size, and hence, a reduction in beam size. This increases the divergence of the beam and compromises the operation of the optical combiner—both reducing the resolution. >2 mm beam diameters are typically required for high resolution. ~30 kHz mirror frequencies are the current state of the art but typically require ~1 mm diameters. [0007] 3. There are also practical limitations on laser modulation rates. ~250-500 MHz are typical practical modulation rates. Higher-speed operation would require complex drive electronics (including high-speed digital to analog converters), and in addition, modulation rates >1 GHz may be limited by the relaxation resonance frequency of the lasers. [0008] 4. The light engine is complex, has a complicated packaging procedure, and is difficult to miniaturize. There are many free-space optical components that are aligned and packaged together.

[0009] Scanning multiple laser beams (“multi-beam scanning”) is one way to alleviate the limitations due to the MEMS mirror fast axis frequency. In multi-beam scanning, multiple beams scan the FOV in parallel, multiplying the number of lines scanned per cycle of the fast axis, and the resolution scales with the number of beams. Conventional techniques to multi-beam scanning involve increasing the number of lasers in the light engine, which further increases the size, complexity, and packaging/assembly difficulty.

[0010] As an example of the limitations of the laser-scanning display approach, achieving a resolution of 2000 pixels×2000 pixels with a state-of-the-art FOV of 36°×36° (50.9° diagonal, state-of-the-art FOVs are ~50° diagonal) at a 120 Hz frame rate involves a fast axis mirror

frequency of about 120 kHz and a laser modulation bandwidth of ~1 GHz (2 ns per pixel addressing time). The mirror frequency is incompatible with a mirror diameter >0.5 mm, and the laser modulation rate would be challenging to achieve due to the complex electronics and the size limitations of head-mounted displays.

[0011] Light detection and ranging (LiDAR) uses light for three-dimensional imaging, often using invisible near infrared (NIR) light with typical wavelengths around 850 nm, 905 nm, or 1550 nm. Miniaturized LiDAR systems are used in automotive sensor and mobile device applications. LiDAR is also used in augmented reality glasses, which map the surroundings of the user such that displayed images/video are in context to those surroundings.

[0012] LiDAR based on laser scanning (where a laser beam is scanned across the scene) has better resolution and contrast than flash methods (where the entire scene is illuminated at once and a pixelated sensor detects the return). In laser scanning LiDAR, the laser beam is modulated (e.g., pulsed/periodically modulated in amplitude or chirped in frequency) and differences in the arrival time/amplitude/frequency of reflected light collected by the LiDAR system enable depth measurements. Laser scanning LiDAR systems are often assemblies of discrete optical components (including lasers, MEMS mirrors, lenses, and photodetectors, like the light engines for display discussed above), posing challenges related to miniaturization and complexity of packaging/assembly. In addition, the MEMS mirrors in such systems limit the resolution, FOV, and frame rate, with faster MEMS mirrors being limited to smaller diameters (resulting in larger beam divergences and lower resolutions).

[0013] Scanning multiple beams can alleviate limitations from the MEMS mirrors, but requires multiple lasers and detection paths, increasing the size and complexity of LiDAR systems based on discrete components. Photonic integrated circuits hold promise for miniaturization and reducing packaging complexity of LiDAR systems. LiDARs based on optical phased arrays become enormously complex for ~1 mm scale beams and may involve hundreds to thousands of phase shifters, and such demonstrations have been limited in their FOV and resolution compared to conventional LiDAR system based on MEMS mirrors.

SUMMARY

[0014] An inventive photonic integrated circuit chip includes a substrate, movable plate, array of lasers, array of waveguides, slab waveguide, grating, and actuator(s). The movable plate is suspended from the substrate and configured to tilt about first and second axes with respect to the substrate. The array of lasers is integrated with the substrate and configured to emit red, blue, and green light; The array of waveguides is integrated with the substrate in optical communication with the array of lasers and guides the red, blue, and green light to the movable plate. The slab waveguide is integrated with the movable plate and guides the red, green, and blue light. The grating is integrated with the movable plate in optical communication with the slab waveguide and emits the red, green, and blue light into free space. And the actuator is operably coupled to the movable plate and tilts the movable plate about the first and second axes so as to scan the red, green, and blue light across at least a portion of a field of view.

[0015] The substrate may form an outer frame around the movable plate, in which case the photonic integrated circuit chip further includes an inner frame suspended from the substrate and configured to tilt about the first axis with respect to the substrate. The movable plate is suspended from the inner frame and configured to tilt about the second axis with respect to the substrate and the inner frame. Such a photonic integrated circuit chip can also include a conductive coil, integrated with the inner frame, to cause two-dimensional rotation of the movable plate via the Lorentz force in response to a current and an applied magnetic field. It can also include inner torsion springs mechanically coupling the movable plate to the inner frame and outer torsion springs mechanically coupling the inner frame to the outer frame.

[0016] The slab waveguide can form the red, green, and blue light into multiple beams of red, green, and blue light, respectively. In this case, the grating can emit the beams of red, green, and

blue light into free space and the actuator and movable plate can scan the beams of red, green, and blue light across different slices of the field of view.

[0017] The grating can be apodized, curved in a plane of the movable plate, and/or configured to emit the red, green, and blue light as collimated beams.

[0018] The grating can include cladding material and first and second layers of grating teeth embedded in the cladding material. The first and second layers of grating teeth diffract first and second portions, respectively, of the red, blue, and green light into free space at first and second angles, respectively, with respect to the movable plate.

[0019] In some cases, the slab waveguide is a first slab waveguide configured to guide first portions of the red, green, and blue light and the grating is a first grating configured to emit first portions of the red, green, and blue light into free space in first directions. In these cases, the photonic integrated circuit chip can also include a second slab waveguide, integrated with the movable plate in optical communication with the array of waveguides, to guide second portions of the red, green, and blue light and a second grating, integrated with the movable plate in optical communication with the second slab waveguide, to emit the second portions of the red, green, and blue light into free space in second directions. The first slab waveguide can overlap the second slab waveguide, and the first grating can overlap the second grating.

[0020] The actuator(s) may include cantilevered piezoelectric actuators coupling the movable plate to the substrate and configured to rotate the movable plate about the first axis and the second axis.

[0021] The array of lasers can also emit infrared light, in which cases the array of waveguides can guide the infrared light to the movable plate, the slab waveguide can guide the infrared light to the grating, and the grating can emit the infrared light into free space. In these cases, the photonic integrated circuit chip can also include at least one photodetector, in optical communication with the grating, to detect returned infrared light from an object illuminated by the infrared light.

[0022] The photonic integrated circuit chip can be part of a light engine for an augmented reality display. This light engine can also include a waveguide combiner, in optical communication with the photonic integrated circuit chip, to direct the red, green, and blue light emitted into free space by the grating, optionally off a mirror, toward an eye of a person viewing the augmented reality display. The array of lasers in such a light engine can emit infrared light, which the array of waveguides can guide to the movable plate. The slab waveguide can guide the infrared light to the grating, which can emit the infrared light into free space. The waveguide combiner can transmit the infrared light onto a scene, and the light engine can include at least one photodetector to detect returned infrared light from the scene.

[0023] An alternative photonic integrated circuit chip can include a substrate, movable plate, array of lasers, array of waveguides, slab waveguide, grating, actuator(s), and photodetector(s). The movable plate is suspended from the substrate and configured to tilt about first and second axes with respect to the substrate. The array of lasers is integrated with the substrate and can emit infrared light. The array of waveguides is integrated with the substrate in optical communication with the array of lasers and can guide the infrared light to the movable plate. The slab waveguide is integrated with the movable plate and can guide the infrared light from the array of waveguides. The grating is integrated with the movable plate in optical communication with the slab waveguide and can emit the infrared light into free space. The actuator is operably coupled to the movable plate and can tilt the movable plate about the first and second axes so as to scan the infrared light across at least a portion of a field of view. And the photodetector is in optical communication with the grating and can detect returned infrared light from an object illuminated by the infrared light. This photonic integrated circuit chip can be part of a LiDAR system, e.g., with a fixed or moving mirror that reflects the infrared light.

[0024] All combinations of the foregoing concepts and additional concepts discussed in greater detail below (provided such concepts are not mutually inconsistent) are part of the inventive subject matter disclosed herein. In particular, all combinations of claimed subject matter appearing at the

end of this disclosure are part of the inventive subject matter disclosed herein. The terminology used herein that also may appear in any disclosure incorporated by reference should be accorded a meaning most consistent with the particular concepts disclosed herein.

Description

BRIEF DESCRIPTIONS OF THE DRAWINGS

[0025] The skilled artisan will understand that the drawings primarily are for illustrative purposes and are not intended to limit the scope of the inventive subject matter described herein. The drawings are not necessarily to scale; in some instances, various aspects of the inventive subject matter disclosed herein may be shown exaggerated or enlarged in the drawings to facilitate an understanding of different features. In the drawings, like reference characters generally refer to like features (e.g., functionally similar and/or structurally similar elements).

[0026] FIG. 1A illustrates an inventive photonic integrated circuit chip, also called a photonic chip. The chip includes integrated red (R), green (G), and blue (B) lasers (possibly flip-chip bonded). The lasers are directly-modulated to encode pixel brightness. Each laser is coupled to a waveguide on the photonic circuit chip, and an example cross-section of a silicon nitride (SiN) waveguide is shown. Each waveguide is routed and coupled to a slab waveguide. Light propagates in the slab and expands in the chip-plane. Light in the slab is then radiated out of the chip plane by a grating with a relatively weak grating strength to achieve a large beam diameter (e.g., about 0.5 mm to about 3.0 mm) with low divergence (e.g., full divergence angles of about 0.005° to about 0.05°). Multiple subsets of RGB waveguides may be coupled to the slab, each corresponding to a different propagation angle in the slab, resulting in multiple independent beams emitted from the chip at various angles.

[0027] The slab and grating are on a movable plate that can rotate along two axes (slow and fast axes), continuously steering the light in two angular dimensions. For example, the movable plate may scan the beam along one axis (the fast axis) at a frequency of about 5-30 kHz and along another axis (the slow axis) at a frequency of about 60 Hz to hundreds of Hz (possibly up to several kHz for Lissajous scanning). The movable plate may produce mechanical deflection on an axis without FOV splitting of about $\pm 5^\circ$ to about $\pm 15^\circ$; with FOV splitting, the mechanical deflection may range from about $\pm 0.5^\circ$ to about $\pm 7.5^\circ$, depending on the amount of FOV splitting.

[0028] FIG. 1B illustrates the photonic integrated circuit chip used as a light engine. The chip is placed in close proximity to the input coupler of an optical combiner (here a waveguide combiner), and the multiple steered beams from the chip are coupled into the combiner, guided via total internal reflection, replicated (via pupil replication elements), and out-coupled toward the eye of the viewer by the output coupler.

[0029] FIG. 1C shows how the photonic chip addresses the FOV of the viewer in slices, each corresponding to an independent, intensity-modulated, steered RGB beam from the photonic chip. Small offsets in the beam angle along the fast axis (with 0 plate deflection) are possible due to the dependence of the grating diffraction angle on the propagation angle in the slab. Offsets between R, G, and B beams within each slice may also occur due to wavelength dependence in the grating diffraction angle and small differences in the slab propagation angle for R, G, and B waveguides within a subset, and in the inset, the effect is exaggerated. These offsets can be compensated for digitally (i.e., by adjustment of the input image data to the light engine).

[0030] FIG. 2A shows a photonic integrated circuit chip with a movable plate based on electromagnetic actuation using a coil. The coil is formed from on-chip metal wires and is defined on the inner frame of a movable plate with a gimbaled design. An applied magnetic field is oriented in the chip plane but diagonal to the principal axes of the inner frame. Oscillating electrical current passed through the coil can cause 2D rotation of the movable plate via the Lorentz force. A slow or

quasi-static oscillation of the current rotates the inner frame (slow axis), and a higher frequency oscillation matching the fast axis resonance frequency rotates the movable plate. Combining both the slow and fast frequency components into the electrical drive signal results in periodic 2D scanning.

[0031] FIG. 2B shows how the magnetic field may be applied by permanent magnets positioned around the photonic chip on a common substrate/carrier (e.g., printed circuit board (PCB)) with the photonic chip.

[0032] FIG. 3A shows a photonic integrated circuit chip with a movable plate and external piezoelectric actuators. The piezoelectric actuators may be formed from a piezoelectric material with electrodes for applying an electric field to the material. Electrical signals applied to the piezoelectric actuators cause vibrations that can couple to mechanical resonances of the fast and slow axes of the movable plate (e.g., for a Lissajous scan pattern).

[0033] FIG. 3B shows a photonic chip with piezoelectric actuators attached to a common carrier/substrate on which the photonic chip is also attached.

[0034] FIG. 3C shows a photonic chip attached on top of piezoelectric actuators.

[0035] FIG. 4 shows a photonic integrated circuit chip with a movable plate and on-chip piezoelectric actuators. The actuators are formed from a piezoelectric material and electrodes deposited on the chip. Similar to FIGS. 3A-3C, electrical signals applied to the actuators lead to vibrations that can excite the slow- and fast-axis mechanical resonances.

[0036] FIG. 5 shows another example of a photonic integrated circuit chip with a movable plate and on-chip piezoelectric actuators. This is a gimbal-less design. Suspended cantilevers with embedded piezoelectric actuators connect to the movable plate via on-chip springs. Electrical signals applied to the actuators modulate the stress in the piezoelectric material, bending the cantilevers. Opposing drive signals applied to different sets of cantilever actuators enables tilting of the platform. Both axes may be operated in resonant or quasi-static mode.

[0037] FIG. 6 shows yet another example of a photonic integrated circuit with a movable plate and on-chip piezoelectric actuators. This is a gimbal-less design. Meandering suspended cantilevers with piezoelectric actuators connect to the movable plate. Electrical signals applied to the actuators cause bending of the cantilevers (similar to FIG. 5), but in this case, by cascading multiple cantilevers (meandering design), the upward/downward motions of each cantilever add up.

[0038] FIG. 7A shows a photonic integrated circuit chip with movable plates and on-chip piezoelectric actuators as in FIG. 5, but with a gimballed design. The fast axis operates on resonance, and the drive signals have a frequency component aligned to the fast axis resonance frequency.

[0039] FIG. 7B shows a photonic chip corresponding to FIG. 6, but with a gimballed design. Again, the fast axis operates on resonance, and the drive signals have a frequency component aligned to the fast axis resonance frequency.

[0040] FIG. 8A shows a top-down illustration of a grating with a waveguide slab and curved grating teeth. An on-axis subset of RGB waveguides couples light into the slab. Light diverges in the slab to a large (possibly millimeter-scale) width and is coupled out of the chip by a grating with curved teeth (the teeth curvature designed to match the phase front of the diverging on-axis beam in the slab and emit a collimated or low divergence beam, e.g., using elliptical grating teeth). This RGB beam addresses a slice of the FOV when scanned by the moving plate.

[0041] FIG. 8B is a top-down illustration of the grating with an off-axis subset of waveguides coupling light into the slab; the light propagates at an angle in the slab, with a corresponding shift in the angle of the emitted beam relative to the on-axis beam. The closely spaced subsets of waveguides form RGB beams.

[0042] FIG. 8C shows cross sections of different grating teeth designs. Grating teeth may be fully etched, partially etched, formed from grating teeth in one or more waveguide layers above the slab waveguide layer (these additional layer(s) being formed from the same material or different

material(s) as the slab), may include a reflector underneath the grating (e.g., a metal reflector or a distributed Bragg reflector), and/or include grating teeth defined by etching the top surface of the cladding of the slab.

[0043] FIG. **9** shows additional grating cross-section possibilities. The grating teeth may be formed in one or more waveguide layers below the slab waveguide layer (these additional layers using the same or different materials as the slab waveguide layer). Alternatively, the grating teeth may be formed in a polymer material on the top surface of the chip and close to the slab waveguide layer (e.g., surface relief gratings, possibly blazed). Additionally, a volume holographic grating may be defined on top of the slab waveguide layer, for example, in a polymer or photographic material. Also, the grating teeth may be laser written in the slab waveguide layer (possibly using ultraviolet light); the laser locally modifying the refractive index of the slab and defining perturbations. The grating teeth can also be formed in a layer parallel to the slab waveguide and situated between a pair of distributed Bragg reflectors.

[0044] FIG. **10A** shows a cross section of an apodized grating. The duty cycle and length of each period in the apodized grating are designed such that: (1) the grating strength increases along the grating with a suitable profile to form a Gaussian beam, and (2) constructive interference occurs between the emissions from all grating periods at the desired diffraction angle (reducing the beam divergence).

[0045] FIG. **10B** shows a top-down illustration of an apodized grating using 2D grating teeth features. Each period of the 2D apodized grating includes a number of patterned features along the transverse axis of the beam in the slab (e.g., using the grating teeth designs in FIGS. **8A-8C** and **9**). The period in the transverse direction is made sufficiently small to avoid coupling light to higher order optical modes of the slab. 2D gratings enable weak grating strengths to be achieved with larger feature sizes since the duty cycle along the transverse direction can be adjusted to weaken the grating.

[0046] FIG. **11A** illustrates a variation of the grating with a slab and curved teeth for interlaced beams. Closely spaced subsets of waveguides coupling light into the slab may have a relatively small angular offset between one another (corresponding to one or a small number of lines in the FOV). The FOV may be addressed by a number of interlaced beams, or alternatively, by a number of sets of interlaced beams, each addressing a slice of the FOV.

[0047] FIG. **11B** is a top-down illustration of the variation of the grating with slab and curved grating teeth with each input waveguide to the slab simultaneously guiding R, G, and B light. This may be accomplished using wavelength multiplexer devices in the photonic integrated circuit (multiplexing light from a set of R, G, and B lasers onto a single waveguide), as in FIG. **35**.

[0048] FIG. **12A** shows a top-down illustration of an alternative slab and grating based on evanescent coupling/leakage of light from waveguides into the slab. If the slab has a higher effective index than the waveguide modes, light from the waveguides leaks into the slab, with the angle of propagation in the slab determined by a phase matching condition. Waveguides are routed toward the slab and curve along an edge of the slab, with the gap between the slab and waveguide determining the coupling strength. The shape of the gap vs. length curve is used to apodize the coupling and form a Gaussian beam in the slab. Multiple waveguides are coupled to the slab and multiple beams are formed in the slab.

[0049] FIG. **12B** shows another view of the alternative slab and grating of FIG. **12A** with light input to a different set of waveguides.

[0050] FIG. **12C** shows a cross-section of the slab and grating of FIG. **12A**, with three waveguides guiding R, G, and B light coupled to the slab. The waveguide widths and/or thicknesses are mismatched to avoid coupling between the waveguides (i.e., the effective indices of optical modes in the three waveguides at R, G, and B wavelengths are different). A subset of RGB waveguides may be coupled to a first facet of the slab, and another subset of RGB waveguides may be coupled to a second facet of the slab. Each subset of waveguides emits an RGB beam into the slab at an

angle relative to the grating teeth as shown in FIGS. 12A and 12B and the resultant two beams emitted from the grating are angled with respect to one another.

[0051] FIG. 12D is a top-down illustration of a simplified slab and grating design with waveguides coupled to a single facet of the slab and the grating teeth angled and aligned to the propagation axis of the beam in the slab. The grating teeth used in this design may be any of those shown in FIGS. 8-10.

[0052] FIG. 13A shows waveguides in multiple layers above, in plane, and below the slab and evanescently coupled to the slab, enable two or more RGB beams to be coupled at a single slab facet. The first subset of waveguides (R1,G1,B1) may have widths and thicknesses designed to couple into the slab at a different angle than the second subset of waveguides (R2,G2,B2), and all waveguides may be designed to have different fundamental mode effective indices at R, G, and B wavelengths (to avoid inter-waveguide coupling).

[0053] FIG. 13B shows three waveguides in the same layer as the slab and that can be coupled to the slab.

[0054] FIG. 13C shows a partially-etched region defined in the gaps between waveguides and slab in FIG. 13B to increase the coupling strength from the waveguides to the slab (potentially enabling larger gaps for a given beam diameter).

[0055] FIG. 14A shows a grating and slab with a lens structure defined in the slab. Light diverging from waveguides coupled to the slab is collimated by the lens structure and then emitted by the grating. The lateral position of the waveguide along the slab input is mapped to angle of the collimated beam in the slab. Multiple beams at various angles may be emitted from the grating.

[0056] FIG. 14B shows another view of the grating and slab of FIG. 14A with light input to a different set of waveguides.

[0057] FIG. 14C The lens structure may be formed from multiple partial etch steps in the waveguide layer.

[0058] FIG. 15A shows a grating design using a tapered waveguide. The taper shape may be nonlinear to reduce the taper length while maintaining high transmission.

[0059] FIG. 15B is a top-down illustration of a grating emitter using a zero-order arrayed waveguide grating.

[0060] FIG. 16 shows a photonic chip with overlapping gratings and slab waveguides on a movable plate to enable FOV splitting along a second direction (here, the fast axis).

[0061] FIG. 17 shows a photonic chip with gratings and slabs in separate layers on a movable plate, with the gratings face away from one another to enable FOV splitting along a second direction.

[0062] FIG. 18 shows a photonic chip with a movable plate that includes a slab in an upper waveguide layer overlapping a grating in a lower set of waveguide layers.

[0063] FIG. 19 shows a photonic chip with four gratings and slabs in four separate layers on a movable plate.

[0064] FIG. 20 shows a photonic chip with two sets of opposing slabs and gratings in two layers on a movable plate.

[0065] FIGS. 21A and 21B show photonic integrated circuit chips with non-rectangular movable plates.

[0066] FIGS. 22A and 22B show photonic integrated circuit chips with non-rectangular movable plates and multiple slabs and gratings.

[0067] FIG. 23A shows top and cross-sectional views of a photonic integrated circuit chip with on- and off-chip reflectors. Steered beams emitted from gratings on the movable plate are reflected toward a reflector defined on the movable plate and then reflected into the input coupler of an optical combiner (e.g., waveguide combiner). The on-chip reflector may be a metal mirror and may be defined over the waveguide slab(s) on the movable plate. The optical beam deflection is three times the mechanical deflection of the plate.

[0068] FIG. 23B shows a side view of the photonic integrated circuit chip of FIG. 23A in

combination with a waveguide combiner.

[0069] FIG. **24** shows a photonic integrated circuit with polarization rotators to convert a transverse-electric (TE) polarization in the waveguides to a transverse-magnetic (TM) polarization. Polarization rotators may be added onto waveguides corresponding to one or more wavelengths (e.g., here, on blue waveguides).

[0070] FIG. **25A** shows an on-chip transmission grating that compensates for grating dispersion (i.e., wavelength-dependent diffraction angle from the grating). The on-chip transmission grating is formed above the grating emitter from one or more patterned dielectric layers and/or grating features etched into the top surface of the cladding.

[0071] FIG. **25B** shows a surface or volume transmission grating defined in a polymer or photographic material may be attached to the surface of the cladding above the grating.

[0072] FIG. **25C** shows a surface or volume reflection grating (e.g., in metals, semiconductors, dielectrics, polymers, or photographic materials) that is aligned above the chip, and the steered beams emitted by the chip reflect off the reflection grating.

[0073] FIG. **26A** shows strain sensors integrated onto the photonic integrated circuit chip (to monitor the deflection of the movable plate). The strain sensors may be positioned near attachment points of torsion springs.

[0074] FIG. **26B** shows the strain sensors, which may be piezoresistive sensors formed from the doped Si device layer of a silicon-on-insulator (SOI) wafer (either etched into wires or with wires defined by doping). The doped Si wires are arranged in a Wheatstone bridge configuration, which is common for strain sensors on MEMS mirrors.

[0075] FIG. **27** shows a variation including actuators on the inner frame and movable plate that apply stress that can compensate for curvature of these parts due to the stresses of the thin films deposited on them (i.e., the waveguide and cladding layers).

[0076] FIG. **28A** shows laser chips aligned to edge couplers on a photonic chip, with the laser chips and photonic chip attached to a common substrate/carrier. The chips may be attached to the substrate face-up or flip-chip bonded.

[0077] FIG. **28B** shows laser chips coupled to edge couplers through small optics (e.g., lenses) that are also attached to the common substrate/carrier.

[0078] FIG. **28C** shows laser array chips used with a photonic chip. A set of individual lasers (with a lithographically defined pitch) may alleviate packaging challenges associated with aligning many laser chips to the photonic integrated circuit chip. In butt-coupling approaches, (e.g., as in FIGS. **28A** and **28C**) and flip-chip bonding (e.g., as in FIG. **1A**), the optical modes at the tips of the edge couplers on the photonic chip should be closely matched to the transverse optical modes of the lasers (for efficient coupling).

[0079] FIG. **29A** shows actuators on a photonic chip used to move the edge couplers in 1D, 2D, or 3D for optimizing the alignment of the edge couplers to the laser chips on a common substrate/carrier with the photonic chip.

[0080] FIG. **29B** shows lasers flip-chip bonded to a photonic chip with actuators to move the edge couplers for aligning the edge couplers to the laser chips.

[0081] FIG. **29C** illustrates an example where tunable waveguide reflectors on a photonic chip are used to reflect light with tuned amplitude, phase, and/or spectrum back into the laser chips (feedback). The tunable on-chip reflectors may include add-drop ring resonators and loop mirrors.

[0082] FIG. **30** illustrates a photonic integrated circuit chip (region near the lasers) with on-chip waveguide-coupled photodetectors (to monitor laser powers) and on-chip temperature sensors (to monitor laser temperatures). The temperature sensors may be formed from thin film metallic wires or doped Si wires; resistance changes with temperature. Power taps (e.g., evanescent directional couplers or multimode interference couplers) couple a small fraction of light from each waveguide to a separate waveguide-coupled photodetector.

[0083] FIG. **31A** illustrates a 1D suspended cantilever actuator with one or more embedded edge

coupler(s) for aligning the edge couplers to lasers in a photonic integrated circuit chip.

[0084] FIG. **31B** shows multiple 1D cantilever actuators connected to a bridge containing the edge couplers in a photonic integrated circuit chip.

[0085] FIG. **31C** shows a 2D actuator (up/down and in-plane/out-of-plane in the illustration) based on cascaded cantilever actuators suitable for use in a photonic integrated circuit chip.

[0086] FIG. **32** shows top-down and cross-section views of tapered, inverse tapered, single layer, and bi-layer edge couplers suitable for use in inventive photonic chips.

[0087] FIG. **33** shows edge couplers made with a two-step etch suitable for use in inventive photonic chips.

[0088] FIG. **34A** illustrates a photonic integrated circuit chip with optical switches, shown as cascaded 1×2 Mach Zehnder interferometer (MZI) switches, connected to each laser.

[0089] FIG. **34B** illustrates a multiplexing scheme for the photonic chip of FIG. **34A**.

[0090] FIG. **35** shows a photonic integrated circuit chip with wavelength multiplexer devices used to multiplex a R, G, and B signal onto each waveguide coupled to a slab; each waveguide then emits an RGB beam into the slab with R, G, and B co-axial.

[0091] FIG. **36** shows a photonic integrated circuit chip where on-chip high-speed optical modulators apply the intensity modulation required to define the brightness of each pixel (instead of direct modulation of the lasers). Here, light from each laser is split into multiple waveguides using on-chip splitter devices; each of these waveguides connects to an on-chip optical intensity modulator followed by routing waveguides connected to the slab(s) and grating(s) for emitting the light. As an example, the optical modulators may be MZI modulators using high-speed phase shifters.

[0092] FIG. **37** illustrates an optical system with a photonic chip and an external MEMS mirror (and no movable plate on the photonic chip).

[0093] FIG. **38** illustrates a system and photonic integrated circuit chip with a 2-axis MEMS mirror integrated onto the photonic integrated circuit chip.

[0094] FIG. **39A** shows an optical system with two 1-axis MEMS mirror chips used to scan the light emitted from the grating(s).

[0095] FIG. **39B** shows an optical system with an external 1-axis MEMS mirror chip and an integrated 1-axis MEMS mirror used to scan the light emitted by the grating(s).

[0096] FIG. **39C** shows an optical system with an external 2-axis MEMS mirror chip positioned above the photonic chip (and on a different carrier/PCB) and used to scan the light emitted by the grating(s).

[0097] FIG. **40** illustrates transmitted and reflected optical signals for (top) time-of-flight pulsed LiDAR, (middle) amplitude modulated coherent wave (AMCW) LiDAR, and (bottom) frequency modulated coherent wave (FMCW) LiDAR.

[0098] FIG. **41** shows a photonic integrated circuit chip for FWCW or AMCW LiDAR with coherent detection.

[0099] FIG. **42** shows an alternative photonic integrated circuit chip for LiDAR. The reflected light is separated by the 2×2 couplers and directly detected by on-chip photodetectors (no coherent detection). This variation is compatible with AMCW LiDAR where extraction of the phase of the amplitude modulation of the reflected light is performed in the electronic domain. This variation is also compatible with pulsed time-of-flight LiDAR, where the arrival time of the reflected pulse is measured.

[0100] FIG. **43A** illustrates a photonic integrated circuit chip being used for LiDAR, with NIR beams emitted by the grating(s) on the movable plate scanned across a scene by the movable plate.

[0101] FIG. **43B** shows a LiDAR system with an off-chip reflector and a reflector integrated onto the movable plate (e.g., a metal mirror above the waveguide slab region of the plate).

[0102] FIG. **44** illustrates alternative LiDAR systems.

[0103] FIG. **45A** shows a photonic integrated circuit chip for simultaneous light engine and LiDAR

functionalities.

[0104] FIGS. **45B** and **45C** show the photonic chip of FIG. **45A** positioned close to the input coupler of an optical combiner.

[0105] FIG. **46** shows a photonic integrated circuit chip and system configuration with gratings and slabs arranged along a curve on the photonic chip.

DETAILED DESCRIPTION

[0106] Laser-scanning optical systems based on photonic integrated circuits can generate multiple modulated, large-diameter, collimated (or low-divergence) steerable beams. Multi-beam scanning has advantages for light engines (the miniature displays in augmented reality glasses) and LiDAR systems compared to systems that scan a single beam. Specifically, scanning the field of view (FOV) by multiple beams in parallel results in faster, higher-resolution scanning that overcomes limitations of conventional single-beam laser scanning systems based on microelectromechanical systems (MEMS) mirrors, where the fast axis frequency of the MEMS mirror fundamentally limits the resolution for a given FOV. Unlike other systems, the laser-scanning optical systems here use photonic integrated circuit chips, also called photonic chips, that emit collimated (or nearly collimated) beams without collimation lenses. Such lens-free, chip-based laser-scanning systems may be much smaller and lighter and have lower packaging complexity compared to laser-scanning optical systems with lenses. Applications of inventive laser-scanning optical systems include light engines for augmented reality glasses, LiDAR systems, and light engines for simultaneous display and LiDAR functionalities.

Light Engines for Augmented Reality Glasses

[0107] FIGS. **1A-1C** illustrate a laser-scanning optical system that includes lasers **120** packaged with a photonic integrated circuit chip **100** (possibly by flip-chip bonding) and coupled to on-chip optical waveguides **130**. The waveguides **130** are coupled to a grating coupler emitter **150** (emitting light out of the chip plane) on a movable plate **140** that can be tilted along two angular axes (using on- or off-chip actuators) for steering of the emitted light. The waveguides **130** are coupled to the grating **150** via a slab waveguide **152** where the light is expanded in the plane of the chip **100**. The light is subsequently radiated by the grating **150**, which has a relatively low grating strength (expanding the light along a second axis)—enabling emission of beams with large diameters (possibly millimeter-scale) with low divergences, without collimation lenses. Coupling multiple waveguides **130** to the slab **152**, each corresponding to a different propagation angle in the slab, results in multiple emitted beams from the grating **150** (each emitted with a different angle). Overall, the result is a set of large, collimated (or low divergence) beams at various angles emitted from the chip **100**, which are continuously steered by rotation of the movable plate.

[0108] As shown in FIG. **1B**, the chip **100** can be placed near the input coupler **106** of an optical combiner (e.g., waveguide combiner **108**) to form a near-eye display system, with the photonic chip **100** functioning as a fully integrated light engine. In the waveguide combiner **108**, the beams are coupled into the waveguide combiner **108**, guided by total internal reflection (TIR), replicated by diffractive or reflective pupil replication elements, and out-coupled toward the eye **11** by an output coupler **110**. Collimated (or low-divergence) beams input to the optical combiner **108** are expanded and output toward the eye **11** forming an image focused at infinity, whereby collimated (or low-divergence) beams with horizontal and vertical angles (θ , ϕ) are focused to position (x , y) on the user's retina. Since the images are focused at infinity, the display system is typically characterized by an angular field of view (FOV). The lasers **120** are intensity modulated to define the brightness of each pixel as the beams are rastered across the FOV of the viewer, with the FOV divided into slices addressed in parallel by the multiple beams.

[0109] The waveguides **130** of the photonic integrated circuits chip **100** may have silicon nitride (SiN) cores **132** with silicon dioxide (SiO₂) cladding **134**, e.g., as shown in the accompanying figures. Other embodiments may use aluminum oxide (Al₂O₃), silicon oxynitride (SiON), doped silica, lithium niobate, or lithium tantalate waveguides with SiO₂ cladding. SiON may

also be used as the cladding rather than SiO₂. Embodiments using polymer waveguides are also possible.

[0110] Grating couplers like the grating **150** are diffractive elements defined in a waveguide that can emit light out of the plane of the chip; they are often defined by etching grooves (“grating teeth”) into the waveguide (slab waveguide **152**). Typically, grating couplers emit light at an angle relative to the normal of the chip (diffraction angle). The strength and dimension of each period may be engineered for radiation of a Gaussian beam (through a design method known as apodization). Furthermore, through apodization and design of the input beam width to the grating **150**, the grating **150** may emit round beams with equal diameters and divergences along the two transverse axes.

[0111] In FIG. **1A**, laser light is coupled to the waveguides of the photonic chip **100** from flip-chip-bonded lasers **120** via edge couplers **128**, which are devices designed to achieve optical mode profiles (at their respective facets) that are well-matched to the transverse optical mode profile of the lasers (for efficient coupling). Each laser **120** is a diode laser mounted on a silicon substrate **110** or silicon device layer of a silicon-on-insulator (SOI) wafer **110** with solder bumps, metal wiring, and/or other mechanical stopper structures **126**. Each laser **120** includes a laser waveguide core **122** with one end covered by a reflective coating **124** and the other (partially reflecting) end exposed to emit red, green, or blue light into free space. Each laser **120** is mated to a corresponding edge coupler **128**, which couples the free-space emission into a corresponding waveguide **130**.

[0112] The waveguides **130**, each guiding red (R), green (G), or blue (B) light, are routed from the edge couplers to the slab waveguide **152**, possibly crossing one another. The waveguides **130** are also routed along on-chip springs **144** connecting the movable plate **140** to the rest of the chip **100**. The waveguides **130** may be arranged into subsets of three waveguides **130** (triplets), each with R, G, and B waveguides, and the intra-subset pitch may be sufficiently small such that the propagation angles of light in the slab coupled from the waveguides in each subset are nearly identical, effectively forming RGB beams in the slab from each subset. Various waveguide subsets may be coupled to the slab forming a set of independent beams in the slab with a variety of propagation angles. The routing waveguides **130** may cross each other in the photonic chip **100**. These crossings may be implemented as in-plane multimode interference crossing devices, tapered waveguide crossings, or over/under-pass crossings using multiple waveguide layers.

[0113] The scan patterns of each beam may be offset from one another along a second axis (fast axis in FIGS. **1A-1C**), which is due to beams propagating at different angles in the slab waveguide and propagating through the grating at an angle, potentially experiencing a different grating period than on-axis light. This effect may be reduced or eliminated through a curved grating design or may be compensated by emitting light from the lasers only when beams are directed to regions of the FOV slices that overlap a common range along the fast axis. Within each FOV slice, some offset between R, G, and B beams is expected along both axes; first, because of the intra-subset pitch leading to small slab propagation angle differences, and second, because the diffraction angle of light emitted from the grating is wavelength dependent. The offsets can be digitally compensated by adjusting the image data input to the display system, for example, by shifting the R, G, and B color components of the input images to align the color components of the final image observed by the viewer. The beams within each FOV slice may slightly overlap with their neighboring slices to avoid gaps in the FOV, the laser intensity may be adjusted in these overlapping regions for uniform brightness of the overall image, and the beams may form an interlaced pattern within these overlapping regions.

[0114] The movable plate **140** in FIG. **1A** is one of multiple examples disclosed herein. Here, an “inner frame” **142** is connected to the rest of the photonic chip **100** by torsion springs **144** (structures mechanically compliant to twisting motion; in a simple example, a long beam that is thin and narrow). The movable plate **140** is connected to the inner frame **142** via two more torsion springs **144**. The inner frame **142** (together with the movable plate **140**) may rotate (tilt) along a

first angular axis relative to the chip **100**, and the movable plate **140** may rotate relative to the inner frame **142** along a second axis. This results in two-dimensional rotation of the movable plate **140** and the grating **150** integrated onto the movable plate **140**. Such configurations are used in MEMS mirrors designs and are referred to as “gimbaled designs.”

[0115] The second axis (rotation of the movable plate **140** within the inner frame **142**) has a higher mechanical resonance frequency than the first axis and may be operated on resonance as the “fast axis” of the light engine, tracing the lines of the FOV. The first axis (rotation of the inner frame **142**) has a lower mechanical resonance frequency and is typically the slow axis. The slow axis may be scanned linearly with a frequency below the resonance to define a uniform line density (quasi-static operation), or operated on resonance, with simultaneous resonant operation along both axes forming a Lissajous scan pattern. The torsion springs **144**, movable plate **140**, and inner frame **142** may be of the same thickness or the torsion springs **144** may be thinner (for increased mechanical compliance).

[0116] Like conventional MEMS mirrors, the movable plate **140**, inner frame **142**, and torsion springs **144** may be formed from the Si layer of a silicon-on-insulator (SOI) wafer with ~10-200 μm thickness. Here, additional dielectric layers are on top (for the waveguide cores **132** and claddings **134**). The gratings **150** may be aligned on the movable plate **140** such that the multiple beams split the FOV along the slow axis (FIG. 1C) or along the fast axis.

[0117] FIGS. 2A and 2B show a photonic chip **200** with a coil **210** defined by on-chip metal wires **212** on the inner frame **142** of a gimbaled design; a second layer of metal wires **216** (passing over/under the first layer) is used for wiring from the inside of the coil **210** to one or more electrical pads **214** outside the inner frame **142**. A substrate or carrier **202**, such as a printed circuit board (PCB), supports the chip **200** and permanent magnets **220** arranged around the chip **200**. These magnets **220** apply a magnetic field oriented in the chip plane and diagonal to the principal axes of the inner frame **142**. Electrical current is passed through the coil **210**, causing the inner frame **142** to move via the Lorentz force. Due to the orientation of the magnetic field, a component of the Lorentz force is aligned to rotate the inner frame **142** (slow axis), and a component of the force can couple to a rotational mode of the movable plate **140** (fast axis). Applying a drive signal composed of a low frequency component (slow axis frequency) and a higher frequency aligned to the resonant frequency of the movable plate **140** (fast axis) enables periodic two-dimensional (2D) scanning of the movable plate along both the fast and slow axes.

[0118] FIGS. 3A-3C and 4 show examples of movable plates with piezoelectric actuators **302**. The piezoelectric actuators are formed from a piezoelectric material **304** with electrodes **306** attached to two surfaces. Electric fields applied to the piezoelectric material **304** via the electrodes **306** can cause the actuator **302** to bend or vibrate. The piezoelectric actuators **302** may be off-chip and attached to a common substrate/carrier **202** along with the photonic chip **100** as shown in FIGS. 3A-3C. In this case, vibration of the piezoelectric actuators **302** can couple to slow- and fast-axis resonances for 2D Lissajous scanning (the movable plate **140** having a gimbaled-design).

Alternatively, the photonic chip may be attached to the top of the piezoelectric actuator(s) **302**. Piezoelectric actuators **302** may also be integrated on the photonic chip **100**, exciting mechanical resonances of gimbaled plates **140** as shown in FIG. 4 (similar to FIGS. 3A-3C).

[0119] On-chip piezoelectric actuators enable more advanced designs (FIGS. 5-7). In FIG. 5, for example, four piezoelectric cantilever actuators **502** are connected to the corners of a movable plate **540** with respective springs **544**. Drive signals applied to the actuators **502** bend the cantilevers. Driving multiple piezoelectric cantilever actuators **502** can rotate the movable plate **540** along two axes (either in resonant or quasi-static mode).

[0120] FIG. 6 shows a photonic chip **600** with meandering suspended piezoelectric cantilever actuators **602** connected to a movable plate **640** with grating emitters. The operation is similar to the operation in FIG. 5, but by cascading multiple cantilevers (meandering design), the upward/downward motions of each cantilever add up.

[0121] FIGS. 7A and 7B show gimbaled photonic chips **700a** and **700b**, respectively, with fast axes that are operated on resonance. The gimbaled photonic chip **700a** in FIG. 7A includes a movable plate **740** with grating emitters that is suspended by torsion springs **744** from an inner frame **742**. The inner frame **742** is coupled, at its corners, to the rest of the photonic chip **700a** by piezoelectric cantilever actuators **702a** like those in FIG. 5. Likewise, the movable plate **740** in the gimbaled photonic chip **700b** of FIG. 7B is suspended by torsion springs **744** from an inner frame **742**. This inner frame **742** is coupled, at its corners, to the rest of the photonic chip **700b** by cascaded piezoelectric cantilever actuators **702b** like those in FIG. 6.

[0122] FIGS. 8-15 show slab and grating designs. In FIGS. 8A-8C, subsets of closely spaced waveguides **130** for red (R), green (G), and blue (B) light connect directly to the slab waveguide **152** (i.e., each waveguide merges with the slab, butt-coupling). The waveguides **130** may be tapered to narrow widths at the interface to enable wide divergences of the emitted beams in the slab **152**. The beams coupled into the slab **152** diverge in the slab **152**. After reaching beam widths close to the desired output beam diameter, the beams are coupled out of the chip by the grating **150**, which has curved grating teeth. The curvature of the grating teeth is designed to match the phase front of the diverging on-axis beam in the slab **152** (for emission of a collimated or low-divergence beam). The grating teeth may have an elliptical shape. The input interface of the slab **152** may be curved (possibly with the same curvature as the grating teeth).

[0123] Off-axis subsets of input waveguides **130** are angled with respect to the center line of the grating **150**/slab **152** and emit angled beams in the slab **152**; the angle increases with the offset. When emitted from the grating **150**, these beams are angled with respect to the on-axis beam. Overall, each beam emitted by the grating **150** addresses a slice of the FOV scanned by the photonic chip, and the offset between subsets of input waveguides **130** determines the spacing of the FOV slices. Unlike other gratings, the grating **150** in FIGS. 8A-8C uses subsets of waveguides **130** for RGB beams where the intra-subset pitch is small and the inter-subset pitch is larger.

[0124] FIGS. 8C and 9 show example cross-sections of grating teeth designs. Such designs can be used with all slab designs disclosed herein and are not exclusive to the designs in FIGS. 8A and 8B. Grating teeth may be fully etched **854a** or partially etched **854b** features in the slab layer **152**. Gratings may further include a back-reflector **856**, i.e., a reflective layer underneath the grating to reflect downward radiation from the grating, which can constructively interfere with the upwards radiation and increase the transmission. The reflector **856** may be a simple metallic mirror layer or a distributed Bragg reflector **952** formed from a stack of thin film layers.

[0125] In addition, grating teeth may be formed from fully etched or partially etched features in one or more waveguide layers above **854c** or below the slab waveguide layer (these additional layer(s) being formed from the same or different materials as the slab layer). Alternatively, these additional layers of grating teeth may be formed from non-waveguide layers **854e**, **950a**, such as metals or Si. In other examples, the grating teeth may be formed in the cladding **854d** or in a polymer material **950b**, **950c** on the top surface of the chip and close to the slab waveguide layer **150** (e.g., surface relief gratings **950b**, possibly having a blazed design **950c**). A volume holographic grating **950d** may be defined on top of the slab waveguide layer **150**, for example, in a polymer or photographic material. Also, the grating **950e** may be laser written in the slab waveguide layer (possibly using ultraviolet light); the laser light locally modifies the refractive index of the slab **150**, defining index perturbations. In another variation, a grating **950f** may be defined in the waveguide layer and distributed Bragg reflectors **952** may be defined above and below the grating.

[0126] Grating apodization makes it possible to emit round, low-divergence beams. FIG. 10A shows an apodized grating **1050a**. The duty cycles and lengths of the grating periods are selected to increase the grating strength along the grating, yielding a Gaussian profile of the emitted beam while ensuring the radiation from each grating period constructively interferes at the required diffraction angle of the grating. Uniform gratings (without apodization) emit beams with an

exponential profile.

[0127] FIG. **10B** shows a grating **1050b** with 2D grating teeth features. Here, the duty cycle of the transverse grating features is designed to weaken the grating strength at the beginning of the grating **1050b**, and the period of the transverse grating features is made small enough to avoid coupling to higher order modes of the slab waveguide **1052**. This approach may allow a larger range of grating strengths than 1D grating features and may allow larger feature sizes to be used.

[0128] FIG. **11A** shows a variation of the grating in FIG. **8** where the subsets of input waveguides **130** are positioned with small lateral and angular offsets from one another to enable interlaced beam scanning. Here, the small lateral offsets lead to small variations in the slab propagation angle of the beams (possibly corresponding to one or a small number of lines of the scan pattern). Interlaced scan patterns increase the line density and hence resolution of the display. Variations where the FOV is scanned by a set of interlaced beams are possible, in addition to variations where the FOV is addressed by sets of interlaced beams, each addressing a slice of the FOV. In the variation shown in FIG. **11B**, each input waveguide **130** may guide and emit R, G, and B light into the slab **152**, enabling smaller offsets between beams in the slab **152** and possibly a higher density of interlaced beams. RGB light may be multiplexed onto waveguides **130** using wavelength multiplexer devices, as in FIG. **35**.

[0129] FIGS. **12A-12D** show a slab and grating design for emitting multiple RGB beams. In this design, waveguides **1230** are routed toward the slab **1252** and curve along an edge of the slab **1252**. As the waveguides **1230** approach the slab **1252**, light evanescently couples/leaks into the slab **1252** since the slab **1252** has a higher effective index than the waveguide modes. The propagation angle in the slab **1252** is determined by a phase-matching condition involving the effective indices of the waveguide mode and slab **1252**. The curvature of the waveguides **1230** defines the gaps between the waveguides **1230** and slab **1252** and hence defines the coupling strength between the waveguides **1230** and the slab **1252** as a function of position along the slab edge. The shape of the curves can be designed to apodize the coupling to the slab **1252** and enable a Gaussian beam in the slab **1252**.

[0130] Unlike other designs, multiple waveguides **1230** are coupled to the slab **1252** and multiple beams are formed in the slab **1252**. As shown in the cross-section in FIG. **12C**, three waveguides **1230** guiding R, G, and B light are coupled to the slab **1252**, and the waveguide widths and/or thicknesses are mismatched to avoid coupling between these waveguides **1230** (i.e., the effective indices of optical modes in the three waveguides at R, G, and B wavelengths are different). A subset of RGB waveguides **1230** may be coupled to a first facet of the slab **1252**, and another subset of RGB waveguides **1230** may be coupled to a second facet of the slab **1252**. Each subset of waveguides **1230** emits an RGB beam into the slab **1252** at an angle relative to the grating **1250**, and the resultant two beams emitted from the grating **1250** are angled with respect to one another.

[0131] FIG. **12D** shows an example of a simplified slab **1252'** and grating **1250'** with waveguides **1230** coupled to a single facet of the slab **1252'** and the grating **1250'** angled and aligned to the propagation axis of the beam in the slab **1252'**. The grating teeth used in the designs in FIGS. **12A-12D** may be like those shown in FIGS. **8-10**.

[0132] FIGS. **13A-13C** show additional variations of waveguides **1330a-1330c** coupled to the slab; waveguides **1330a** may be in layers above and below the slab **152**; R, G, and B waveguides **1330b** may all be in the same waveguide layer as the slab **152**; and a partially-etched layer may be in the gaps between the waveguides **1330c** and the slab **152** (to increase the gap required to achieve a certain coupling strength).

[0133] FIGS. **14A-14C**, **15A**, and **15B** show additional grating and slab designs. Briefly, FIGS. **14A-14C** shows a design where a lens structure **1454** is included in the slab **1452** to collimate diverging beams in the slab **1452**, with a grating **1450** to couple the collimated beams out of the photonic chip. The input waveguides **130** to the slab are arranged in subsets of closely spaced waveguides to emit RGB beams.

[0134] FIG. 15A shows a design where light from a single input waveguide **1530** is expanded by a linear or nonlinear waveguide taper **1554** prior to being output by a grating **1550**. FIG. 15B shows a design where a zero-order arrayed waveguide grating (AWG) is used to generate a wide beam in a slab **1552**, which is subsequently output by a grating **1550**. A star coupler **1560** splits light from an input waveguide **1530** into a number of output waveguides **1562**, which are routed toward the slab waveguide **1552**. These routed waveguides **1562** have equal optical path lengths and end in tapers **1556** that increase in width at the input to the slab **1552**. By matching the optical path lengths, the light from the waveguides **1562** at the input to the slab **1552** is in phase, and due to the tapers **1556**, a collimated beam with a large width can be injected into the slab **1552**. The coupling strength of the star coupler **1560** to the array of waveguides **1562** can be designed to achieve an approximately Gaussian profile of the beam in the slab **1552**. A grating **1550** in the slab layer couples the light out of the slab **1552**. This grating **1550** may be positioned very close to the beginning of the slab **1552**. This design couples one RGB beam out of the slab **1552** and can reduce the size of the overall device compared to other grating designs but may be more sensitive to fabrication variation/error. The designs in FIGS. 14A-14C may emit multiple RGB beams, whereas the designs in FIGS. 15A, and 15B each only emit a single RGB beam.

[0135] FIGS. 16-20 show variations of the photonic integrated circuit design where multiple slabs and gratings are integrated onto the movable plate. Integrating multiple gratings onto the movable plate enables FOV slicing along a second direction, i.e., the FOV is separated into a matrix of slices, each addressed by an independent beam emitted from one of the gratings. One consideration in such designs is the size of the movable plate. Large plates have low fast-axis frequencies, which may impose resolution limitations, and can suffer from increased dynamic deformation (i.e., small deformations of the plate during motion that can diminish the emitted beam quality when larger than a tenth of the wavelength). Also, the slabs add overhead to the plate size, and the plates are significantly larger than the millimeter-scale gratings. Further doubling the plate length or width to add a second grating may be impractical given the above considerations.

[0136] FIGS. 16-20 show different variations of multiple slabs and gratings in different waveguide layers that overlap one another. This enables integration of multiple gratings in a smaller plate area than might otherwise be possible with multiple gratings in a single waveguide layer. Overlapping two slabs in different layers is one approach with the advantage that the grating emissions do not interact with waveguide layers above, but the plate length or width still increases by the length of a grating. Alternatively, the slabs and gratings of upper layers may overlap the gratings of lower layers, with a small reflection expected from the slab above each grating. Gratings above other gratings may be designed to avoid significant diffraction of the beams passing through them.

[0137] FIG. 16 shows gratings **150**, **150'** and slabs **152**, **152'** on a movable plate **140**. The gratings **150**, **150'** and slabs **152**, **152'** are in two or more sets of waveguide layers directly overlapping. This enables FOV splitting along a second direction (here, the fast axis).

[0138] FIG. 17 shows other gratings **150**, **1750** and slabs **152**, **1752** on a movable plate **140**. Here, the slabs **152**, **1752** in two separate layers overlap, while each grating **150**, **1750** only has the cladding above it. The two gratings **150**, **1750** face away from one another. As in FIG. 16, FOV splitting is enabled along a second direction.

[0139] FIG. 18 shows still other gratings **1850**, **1850'** and slabs **1852**, **1852'** on a movable plate. Here, the slab **1852'** of a top waveguide layer overlaps the grating **1850** of a bottom set of waveguide layers. This is another example of FOV splitting along a second direction.

[0140] FIG. 19 shows yet more gratings **150**, **150'**, **150''**, **150'''** and slabs **152**, **152'**, **152''**, **152'''** on a movable plate **140**. Four gratings **150**, **150'**, **150''**, **150'''** in four sets of layers are arranged such that light from the bottom three gratings **150**, **150''**, **150'''** passes through only slabs **152**, **152'** in the upper layers. FOV splitting along a second direction is enabled by this design.

[0141] FIG. 20 shows another variation of gratings **2050**, **2050'**, **2050''**, **2050'''** and slabs **2052**, **2052'**, **2052''**, **2052'''** on the movable plate. Two gratings **2050**, **2050'** and slabs **2052**, **2052'** are

defined in a top set of waveguide layers with the gratings **2050**, **2050'** facing one another. Two gratings **2050''**, **2050'''** and slabs **2052''**, **2052'''** are defined in a bottom set of waveguide layers with the gratings **2050''**, **2050'''** facing away from one another. Light emitted from the bottom gratings **2050''**, **2050'''** passes through the top slabs **2052**, **2052'**. This enables FOV splitting along a second direction.

[0142] Since in-plane beam expansion occurs in the slab **152**, the slab **152** may be non-rectangular and may taper. FIGS. **21A**, **21B**, **22A**, and **22B** show examples where the movable plate may also be non-rectangular, reducing the mass and area of the plate for higher fast-axis frequencies and less dynamic deformation. The connection point of torsion springs to the movable plate may be aligned to the center of mass of the plate to achieve rotational motion without out-of-plane up/down motion.

[0143] FIGS. **21A** and **21B** show photonic integrated circuit chips **2100a** and **2100b**, respectively, with non-rectangular movable plates **2140a** and **2140b**. Since light expands in the waveguide slab **152**, starting from a small width and ending with a larger width, the slab **152** can be tapered. Correspondingly, the movable plates **2140a** and **2140b** can have non-rectangular shapes to reduce their overall mass and area. The torsion springs **144** may be positioned along an axis corresponding to the center of mass of the non-rectangular movable plate **2140a**, **2140b**. The grating **150** and slab **152** may be aligned relative to the non-rectangular movable plate **2140a**, **2140b** for FOV splitting along the fast axis (FIG. **21A**) or along the slow axis (FIG. **21B**).

[0144] FIGS. **22A** and **22B** show photonic integrated circuit variations **2200a** and **2200b**, respectively, with non-rectangular movable plates **2240a**, **2240b**, each with multiple slabs **152**, **2252** and gratings **150**, **2250**. The non-rectangular movable plate **2240a** in FIG. **22A** is hexagonal and features gratings **150**, **2250** that face each other. The non-rectangular movable plate **2240b** in FIG. **22B** is bowtie-shaped and features gratings **150**, **2250** that face away from each other.

[0145] FIGS. **23A** and **23B** show a variation of the photonic integrated circuit chip **2300** and system including an on-chip reflector **2302** on the movable plate **140** and an off-chip reflector **2304** mounted on the waveguide combiner **108**. Here, the on-chip reflector **2302** may be a thin-film metallic mirror deposited over the cladding **134** of the slab **152** region of the plate **140**. Beams emitted from the grating **150** are directed back toward the on-chip reflector **2302** by the off-chip reflector **2304** and subsequently directed toward the input coupler **106** of the waveguide combiner **108**. Since the grating **150** and on-chip mirror **2302** both tilt with the movable plate **140**, the optical deflection angle of the beam is three times the mechanical deflection of the movable plate **140**. This reduces the mechanical deflection of the movable plate **140** for a given FOV. For comparison, in systems with a conventional MEMS mirror with a fixed incident beam, the optical deflection angle is twice the mechanical deflection of the MEMS mirror. Conversely, for a movable plate with integrated slabs and gratings but without on- and off-chip reflectors, the optical deflection angle is equal to the mechanical deflection angle.

[0146] FIG. **24** shows a photonic integrated circuit chip **2400** with routing waveguides and crossings **2402** as well as waveguide polarization rotators **2404** integrated into waveguides **130** that guide one or more of the colors. Laser diode emission is typically transverse-electric (TE) polarized, while some grating designs benefit from transverse-magnetic (TM) polarized light. The waveguide polarization rotators **2404** transform TE-polarized light into TM-polarized light for higher diffraction efficiency from these gratings or reducing diffraction angle differences between shorter wavelength (e.g., blue) emitted beams and longer wavelength (e.g., green, red) emitted beams.

[0147] FIGS. **25A-25C** show variations of the photonic integrated circuit chip for compensation of grating dispersion (i.e., the variation of diffraction angle with wavelength). Over the 1-2 nm typical linewidths of typical laser diodes, the diffraction angle of typical on-chip grating couplers shifts on the order of 0.1°, which limits the divergence of the emitted beam. High-resolution laser-scanning light engines have lower divergence (e.g., a 2000×2000-pixel resolution and 35°×35° FOV, which

implies a divergence $<35^\circ/2000=$) 0.0175° . Narrower linewidth lasers may be used to avoid this limitation (and may be achieved using optical feedback from on-chip tunable reflectors, as in FIG. 29C), but narrow linewidths can cause speckle in optical combiners (due to long laser coherence lengths).

[0148] FIGS. 25A-25C show add-ons to the photonic circuit chip that compensate the grating dispersion. The dispersion of transmission and reflection gratings can be opposite to the dispersion of waveguide-coupled gratings, i.e., the diffraction angle (relative to the chip normal) of a waveguide-coupled grating typically decreases with increasing wavelength, while the diffraction angle of a transmission grating can increase with increasing wavelength. With appropriate design of a transmission grating, the magnitude of the dispersion may match or be similar to the grating emitter dispersion, reducing the beam divergence.

[0149] FIG. 25A shows examples where a transmission grating 2502 is defined on the photonic chip (above the grating emitter) with one or more dielectric, metal, or semiconductor layers and/or grating features 2504 etched into the top surface of the cladding 134. FIG. 25B shows an example where a transmission grating 2506 may be attached to the top surface of the movable plate 140 above the grating emitter 150, and the transmission grating 2506 may be a surface relief or volume holographic grating defined in a polymer or photographic material. FIG. 25C shows an example where a reflection grating 2508 may be positioned above the photonic chip, with a design and orientation designed for cancellation of dispersion of the grating emitter 150.

[0150] FIGS. 26A and 26B show a photonic integrated circuit chip 2600 with strain sensors 2610 positioned at the connection points of torsion springs 144. For example, each strain sensor 2610 can be implemented as doped silicon 2612 arranged in a Wheatstone bridge with conductive vias 2614 at the corners that connect the doped silicon 2612 to metal wiring and pads (not shown). Each strain sensor 2610 can be made of a P++, N++, P, or N doped region of a silicon layer, e.g., in a portion 2612 of an SOI wafer surrounded by cladding 134 as shown at left in FIG. 26B or in a portion 2612' of the silicon substrate 110 of the photonic chip. In operation, the strain sensors 2610 provide a measure of the deflection of the movable plate 140 for control of the system.

[0151] FIG. 27 illustrates a photonic integrated circuit chip 2700 with thermal actuators 2702 and/or piezoelectric actuators 2702' embedded in the cladding 134 on the inner frame 142 and on the movable plate 140. The actuators 2702, 2702' compensate for out-of-plane curvature caused by material stresses from the deposited dielectric layers on the chip 2700 (for waveguide cores 132 and cladding 134). The piezoelectric actuators 2702' may be formed of piezoelectric material 304 sandwiched between metal layers 306 for applying an electric field to the piezoelectric material 304 as shown at lower right in FIG. 27. The thermal actuators 2702 may include a metal layer 2708 near a (resistive) heater 2710 that heats metal layer 2708, causing the metal layer 2708 to expand and/or bend.

[0152] As described above, flip-chip bonding can be used to couple laser chips 120 to photonic integrated circuit chips 100. Other possibilities are shown in FIGS. 28A-28C. In FIG. 28A, laser chips 120 are aligned to edge couplers 2804 on a photonic chip 2800a, and both the laser chips 120 and the photonic chip 2800a are attached to a common substrate/carrier 2802a. The laser chips 120 and the photonic chip 2800a may be attached to the substrate 2802a face-up or flip-chip bonded. In FIG. 28B, the laser chips 120 are coupled to edge couplers in a photonic chip 2800b through small optics 2804 (e.g., lenses) that are also attached to the common substrate/carrier 2802b. FIG. 28C shows a photonic chip 2800c with laser array chips 2820 (with a lithographically defined pitch) instead of individual laser chips to alleviate packaging challenges associated with aligning many laser chips to the photonic integrated circuit chip 2800c. In butt-coupling approaches (e.g., FIGS. 1, 28A, and 28C), efficient optical coupling involves matching of the optical waveguide mode of each edge coupler of the photonic chip to the waveguide mode of the corresponding laser chip.

[0153] FIGS. 29A-29C show additional possibilities for coupling laser chips to a photonic integrated circuit chip. In FIG. 29A, the laser chips 120 and photonic chip 2900a are mated to a

common carrier/substrate **2902**. Edge couplers **2904** on 1D, 2D, or 3D actuators, all integrated with the photonic chip **2900a**, couple light from the laser chips **120** into waveguides **130** integrated with the photonic chip **2900a**. In FIG. **29B**, the laser chips **120** and the actuator-mounted edge couplers **2904** are integrated directly with the photonic chip **2900b** itself. In both cases, the actuators may be used to translate and/or tilt the input edge couplers **2904** to optimize the alignment of the edge couplers **2904** to the lasers **120**.

[0154] FIG. **29C** shows a variation of the laser coupling where tunable waveguide reflectors **2910** on the photonic integrated circuit chip **2900c** provide feedback to the laser chips **120**. The tunable reflectors **2910** may be tunable in amplitude, reflection spectrum, and/or phase. An example of such a reflector **2910** is an add-drop ring resonator **2912** with a waveguide loop mirror **2916** at the drop port; tuning the ring resonator **2912** changes the wavelength of maximum reflectivity and tuning the phase shifter **2914** between the edge coupler **2902'** and ring resonator **2912** tunes the phase of the reflected light. Tunable feedback to the corresponding laser **120** enables tuning of the wavelength, linewidth, and stability of that laser **120**.

[0155] In photonic integrated circuit chips with flip-chip bonded lasers (e.g., as in FIG. **1A**), the lasers may be placed in trenches defined in the wafer substrate surface. These trenches may have pillars defined in them that serve as mechanical stops for accurate height alignment of the laser core and edge coupler when placing the chip (self-alignment). These trenches may also contain on-chip wiring and solder bumps for soldering the laser in place and applying electrical signals to the laser.

[0156] FIG. **30** shows a photonic integrated circuit chip **3000** with a laser coupling portion that uses waveguide power taps **3004** (e.g., multimode interference couplers or evanescent directional couplers) to couple small fractions of the light in the waveguides **130** to respective on-chip waveguide-coupled photodetectors **3006** for monitoring the power and intensity modulation of the lasers **120**. Such a waveguide-coupled photodetector **3006** may include a waveguide (e.g., made of SiN or Al_{0.2}O_{0.3}) passing over a patch or mesa of doped Si, wherein a PIN or PN junction is defined. FIG. **30** also shows that temperature sensors **3008** may be defined on the photonic integrated circuit chip **3000** in close proximity to the lasers **120**. Temperature sensors **3008** can be used for ensuring a stable operating temperature (and wavelength) of the lasers **120**. Temperature sensors **3008** may be formed from thin film metal wires or doped Si wires formed on the photonic chip (in both cases, the electrical resistance changes with temperature, providing a measure of temperature).

[0157] FIGS. **31A-31C** show examples of actuators suitable for use in inventive photonic chips. Such actuators may be based on electro-thermal or piezoelectric actuation and may include suspended cantilevers containing the edge couplers or cascaded cantilever actuators for linear translation of a bridge containing the edge couplers. Cascaded cantilever actuators can perform large vertical and in-plane translation. An actuator formed from a single cantilever with a waveguide can be used for aligning to a laser and a fiber. Following alignment using the actuator, epoxy may be applied to the actuator to fix its position.

[0158] FIG. **31A** illustrates a 1D suspended cantilever actuator **3100a** with one or more embedded edge coupler(s) **3102** for aligning the edge couplers **3102** to lasers (not shown) in or coupled to an inventive photonic chip. The 1D suspended cantilever actuator **3100a** includes a suspended cantilever **3104** with an actuator **3106** that bends the cantilever **3104**, tilting and/or translating the actuator **3106** out of the chip plane. The actuator **3100a** may be a piezoelectric or electrothermal actuator as described above with respect to FIG. **27**.

[0159] FIG. **31B** shows another 1D suspended cantilever actuators **3100b** with multiple suspended cantilevers **3104**, each with its own actuator **3106**, coupled to a bridge with extensions **3108**, each of which contains one or more edge couplers **3102**. The multiple cantilevers **3106** may increase the robustness of the suspended structure.

[0160] FIG. **31C** shows a 2D actuator **3100c** (up/down and in-plane/out-of-plane in the illustration)

based on cascaded cantilever actuators **3112** (e.g., using piezoelectric or electrothermal mechanisms like those described above). One or more actuators for each actuation direction are connected to a suspended bridge **3114** with extensions **3108** containing the edge coupler(s) **3102**. Cascaded cantilever actuators **3112** enable larger translations than simple cantilever actuators, and additionally, can be designed to translate without tilting.

[0161] FIGS. **32** and **33** show variations of edge couplers that may be used to couple light from the laser chips to the photonic integrated circuit chip. The edge couplers are designed such that the optical mode at the tip of the edge coupler closely matches the transverse optical mode of the corresponding laser chip. FIGS. **32** and **33** show single-layer and bi-layer edge couplers in addition to tapered and inverse-tapered designs.

[0162] More specifically, FIG. **32** shows a single-layer tapered edge coupler **3202**, single-layer inverse tapered edge coupler **3200**, bi-layer tapered edge coupler **3206**, and bi-layer inverse tapered edge coupler **3204**. Single-layer edge couplers **3200**, **3202** (*top*) have waveguide cores **132**, **3232'** that increase or decrease the width of the waveguide **130** as it approaches the facet of the photonic chip (and are terminated with a blunt tip where light is coupled into the waveguide **130** at the chip facet). Bi-layer edge couplers **3204**, **3206** (*bottom*) include inter-layer transitions where overlapping tapers between two waveguide layers **132**, **3232''** enable light to transfer between the layers; a thicker waveguide **132** (useful for compact on-chip photonic devices and dense waveguide routing) is coupled to a thinner waveguide layer **3232''** with lower optical confinement (more ideal for engineering the optical mode for input coupling from lasers). Both tapered and inverse tapered edge couplers can function as the input edge couplers (for laser coupling) of an inventive photonic chip.

[0163] FIG. **33** shows edge couplers **3302**, **3304** using partial and full etch steps (two-step etch) of a single waveguide layer **132** to define a thin waveguide for coupling light on the chip and a thicker waveguide **3332** for compact photonic devices and dense waveguide routing away from the facet.

[0164] FIGS. **34-36** show additional photonic integrated circuit variations. In FIGS. **34A** and **34B**, on-chip optical switches **3410** route light from the waveguides **130** coupled to the lasers **120** to a larger number of waveguides **130** that couple to the slabs **152** and gratings **150**. This enables a scheme where the FOV is addressed in sub-frames where the slow-axis deflection can decrease by a factor of the (number of subsets of waveguides input to slab/number of lasers per color) at the expense of a correspondingly higher slow axis frequency.

[0165] More specifically, FIG. **34A** illustrates a photonic integrated circuit chip **3400** with optical switches **3410**, shown as cascaded 1×2 Mach Zehnder interferometers (MZIs) **3402**, connected to each laser **120**. Each MZI **3402** has a phase shifter in one of the arms (e.g., thermo-optic phase shifter with a thin film heater above the waveguide), and the phase shifter is driven with an electrical signal to switch the input between the two outputs. The number of waveguide subsets coupled to the slab **152** is larger than the number of lasers **120** per color, and the switches **3410** route laser light to the waveguides **130**.

[0166] FIG. **34B** illustrates the multiplexing scheme for the photonic chip of FIG. **34A**. Each frame of the display is divided into a number of sequential sub-frames, each corresponding to a set of waveguides coupled to the slab selected by the optical switches. The frame is complete after all waveguides have been addressed and their respective output beams have scanned their FOV slices.

[0167] FIG. **35** shows a photonic chip **3500** that uses waveguide crossing **3502** and on-chip RGB wavelength multiplexers **3504** to combine light from one set of R, G, and B lasers **120** onto each waveguide **3530** coupled to the slab waveguide **152**. The emitted light from each waveguide **3530** coupled to the slab waveguide **152** includes co-axial R, G, and B beams, in contrast to FIG. **1A**, which uses a separate waveguide **130** for each R, G, and B beam. This approach may reduce or eliminate the offset between R, G, and B beams in FIGS. **1A-1C**, at the expense of additional photonic integrated circuit complexity and optical loss. The wavelength multiplexers **3504** may be arrayed waveguide gratings, evanescent directional couplers, ring resonators, or echelle gratings.

[0168] FIG. 36 shows a photonic chip 3600 with on-chip splitting devices (power splitters) 3602 that split the light from each laser 120 into a number of waveguides 130, each connected to an on-chip high-speed intensity modulator 3604, which modulates the intensity of each beam, rather than directly modulating the lasers 120. The optical splitting devices 3602 may be cascaded 1×2 multimode interferometer or evanescent directional coupler splitters. These intensity modulators 3604 may operate at modulation rates of about 100-500 MHz. The modulators may be formed from microelectromechanical (MEMS) phase shifter devices, semiconductors such as gallium nitride (GaN), or electro-optic materials such as barium titanate (BTO), lithium niobate, or lithium tantalate. Additional direct modulation of the lasers may be used to turn off the laser in empty regions of the image.

[0169] FIGS. 37 and 38 show photonic integrated circuit chips 3700 and 3800, respectively, where the slab(s) 152 and grating(s) 150 are not on a movable plate. Instead, each system has a two-axis MEMS mirror (either integrated onto the photonic chip or on a separate chip) that scans the beams. For the case of an integrated MEMS mirror, an external reflector above the chip redirects the beam(s) emitted by the photonic chip to the integrated MEMS mirror.

[0170] In FIG. 37, the beam(s) emitted by the grating(s) 150 on the photonic chip 3700 are reflected off an external reflector 3702 (e.g., a mirror or grating) and directed towards a 2-axis MEMS mirror chip 3706. Both the photonic chip 3700 and the 2-axis MEMS mirror chip 3706 are attached to a common carrier/substrate 202. The 2-axis MEMS mirror chip 3706 includes an on-chip reflector (MEMS mirror) 3704, such as a metal mirror, mounted on a movable plate 3740 suspended from an inner frame 3742 by torsion springs 3744. This inner frame 3742 is itself suspended from the MEMS mirror chip 3706 by another set of torsion springs 3744 to provide two-axis gimbaled motion of the on-chip reflector 3704.

[0171] In operation, the MEMS mirror 3706 scans and directs the beam(s) to the input coupler of an optical combiner. Beams emitted at different angles from the grating 150 shift laterally as they propagate through the system. This limits the range of beam angles that may be used if multiple beams are emitted from the photonic chip 3700. Differences in the diffraction angles of R, G, and B beams may be corrected, e.g., by including spacers 3710 and wavelength-selective reflectors 3712 (e.g., dichroic mirrors) that are tilted with respect to one another (top-right) in the off-chip reflector 3702. The spacers 3710 correct for lateral shifts, and the tilt angles of the reflectors 3712 correct for differences in the diffraction angle of R, G, and B light.

[0172] FIG. 38 illustrates a photonic integrated circuit chip 3800 with an integrated 2-axis MEMS mirror 3806. The 2-axis MEMS mirror 3806 includes an on-chip reflector 3804, such as a metal mirror, mounted on a movable plate 3840 suspended from an inner frame 3842 by torsion springs 3844. This inner frame 3842 is itself suspended from the photonic chip 3800 by another set of torsion springs 3844 to provide two-axis gimbaled motion of the on-chip reflector 3804. Similar to the photonic chip 3700 in FIG. 37, light emitted by the grating(s) 150 is reflected by an external reflector 3702 toward the on-chip MEMS mirror 3804. The MEMS mirror 3804 scans the beam(s) and directs them to the input coupler 106 of the waveguide combiner 108. Similar to FIG. 37, the off-chip reflector 3702 may be configured to compensate for differences in the R, G, and B emission angles from the grating(s) 150.

[0173] FIGS. 39A-39C show light engines based on these concepts together with a waveguide combiner 108. Grating dispersion and diffraction angle differences between R, G, and B light may lead to significant lateral offsets between R, G, and B beams before reaching the input coupler 106 of the waveguide combiner 108. The off-chip reflector may align R, G, and B beams to compensate for wavelength-dependent dispersion and/or diffraction.

[0174] The light engine in FIG. 39A includes a photonic integrated circuit chip 3900a without a movable plate. This photonic chip 3900a mounted next to a single-axis MEMS mirror chip 3906a on a common carrier/PCB 202. The grating 150 couples beams out of the photonic chip 3900a to another single-axis MEMS mirror chip 3902, which scans the beams in a first direction (e.g., into

and out of the plane of the figure) across the single-axis MEMS mirror chip **3906a**, which scans the beams in a second direction (e.g., in the plane of the figure; orthogonal to the first direction). The input coupler **106** receives the scanned beams, which propagate through the waveguide combiner **108** to the output coupler **110**.

[0175] The light engine in FIG. **39B** includes a photonic integrated circuit chip **3900b** without a movable plate and with an integrated single-axis MEMS mirror **3906b**. The grating **150** couples beams out of the photonic chip **3900b** to another single-axis MEMS mirror chip **3902**, which scans the beams in a first direction (e.g., into and out of the plane of the figure) across the single-axis MEMS mirror on the photonic chip **3906b**, which scans the beams in a second direction (e.g., in the plane of the figure; orthogonal to the first direction). The input coupler **106** receives the scanned beams, which propagate through the waveguide combiner **108** to the output coupler **110**.

[0176] The light engine in FIG. **39C** includes a photonic integrated circuit chip **3900c** without a movable plate or an integrated single-axis MEMS mirror. In this light engine, the grating **150** couples beams out of the photonic chip **3900c** to a two-axis MEMS mirror chip **3910**, which scans the beams in two directions across the input coupler **106**. The input coupler **106** couples the scanned beams into the waveguide combiner **108**, which guides them to the output coupler **110**.

LiDAR Systems

[0177] FIG. **40** shows three LiDAR techniques compatible with laser scanning. In time-of-flight pulsed LiDAR (top), at each position of the beam, an optical pulse is emitted and the time delay in receiving a reflected (echo) pulse from an object enables calculation of the object's depth or distance from the optical pulse source. In amplitude-modulated coherent-wave (AMCW) LiDAR (middle), the amplitude of the beam is modulated periodically, and the phase difference of the amplitude modulation between the transmitted light and received reflected light enables calculation of the object's depth or distance. Lastly, in frequency-modulated coherent-wave (FMCW) LiDAR (bottom), the optical frequency of the laser is modulated periodically, and the frequency difference between transmitted light and received reflected light is used to calculate the depth of or distance to an object. With each method, as the laser beam is scanned across a scene, depth information can be extracted for every beam position.

[0178] FMCW LiDAR is often implemented by interfering the received reflected light and a portion of the laser light tapped off before transmission (referred to as the local oscillator (LO) beam). The interference may be performed with a beam splitter or other 2×2 optical coupler, and the two resultant outputs are detected with balanced photodetectors (coherent detection). An optical frequency difference between the LO and the reflected light leads to a beat signal in the photocurrent output of the balanced photodetectors, and the beat frequency is used to calculate the depth. AMCW may be performed with coherent detection or by detecting the reflected light with a single photodetector and extracting the phase shift of the amplitude modulation with electronic circuits. Time-of-flight pulsed LiDAR is performed by detecting the reflected pulse with a single photodetector.

[0179] FIG. **41** shows a photonic integrated circuit chip **4100** for FMCW or AMCW LiDAR with coherent detection (it is especially well-suited for FMCW LiDAR). Like the light engines for displays, this photonic chip **4100** has one or more slabs **152** and gratings **150** integrated onto a movable plate **140**. Here, the photonic chip **4100** has operating wavelengths in the near-infrared (NIR) portion of the electromagnetic spectrum (e.g., wavelengths of 850 nm, 905 nm, 940 nm, and 1550 nm), so NIR lasers **4120** are flip-chip bonded onto or otherwise co-packaged with the photonic chip **4100**. The wavelengths of the lasers **4120** do not have to be identical, and in some embodiments, the wavelengths of the lasers **4120** may be different to limit crosstalk between reflected beams received by the photonic chip. In embodiments using wavelengths longer than about 1100 nm (e.g., 1550 nm), Si (e.g., using the Si device layer of a silicon-on-insulator wafer) may be used as the waveguide core, waveguide slab, and/or grating materials.

[0180] The NIR beams emitted by the NIR lasers **4120** can be modulated in amplitude or frequency

by driving the NIR lasers **4120** directly or with external modulators (not shown). Waveguides **4130** guide the modulated NIR beams from the NIR lasers **4120** to tunable on-chip reflectors **4110** (e.g., like those described above with respect to FIG. **29C**), which are used to provide optical feedback to the NIR lasers **4120** for achieving narrow linewidths. The on-chip reflectors **4110** reflect portions of the modulated beams back to the NIR lasers **4120**. This helps to ensure that the NIR lasers' coherence lengths are significantly larger than twice the maximum depth measured. They also couple portions of the modulated beams to frequency discriminators **4112** that monitor the beams' optical wavelength(s), and, with control electronics (not shown), enable stabilization and tracking of the beams' optical frequency.

[0181] A 1×2 coupler couples off a portion of the modulated beam from each NIR laser **4120** to a separate set of waveguides to form the local oscillators (LOs). The waveguides **4130** guide the rest of each modulated beam to the grating **150** via the slab waveguide **152** (with 2×2 couplers splitting portions of the beams off to photodetectors for power monitoring). As explained in greater detail below with respect to FIGS. **43A** and **43B**, the grating(s) **150** emit collimated (or low-divergence) beams, which are scanned across the scene by the movable plate **140**. Beams reflected or scattered from the scene follow the same path backwards and are coupled back into the waveguides **4130** via the grating(s) **150** from which they were emitted. A portion of the reflected light in each waveguide **4130** is coupled into a separate set of waveguides by 2×2 couplers, which interfere the reflected light with the LOs. On-chip balanced photodetectors **4114** coupled to the outputs of the 2×2 couplers detect this interference. The photodetectors **4114** may be waveguide-coupled Si photodetectors, which can detect light at wavelengths less than about 1000 nm. For wavelengths longer than about 1100 nm (e.g., 1550 nm), the photodetectors **4114** may be waveguide-coupled germanium photodetectors.

[0182] FIG. **42** shows a photonic integrated circuit chip **4200** for time-of-flight pulsed LiDAR or AMCW LiDAR (without coherent detection). Like the photonic integrated circuit chip **4100** in FIG. **41**, the photonic integrated circuit chip **4200** in FIG. **42** features directly modulated NIR lasers **4120** that generate (amplitude- or pulse-) modulated NIR beams that are emitted from a grating **150** and scanned across the scene with a movable plate **140**. The grating **150** couples beams reflected and/or scattered from the scene back into the waveguides **4130**, which route them via 2×2 couplers to photodetectors **4202** that detect the reflected and/or scattered beams. (The 2×2 couplers **4200** also couple portions of the outgoing beams to photodetectors **4204** that monitor the beams' output power.) The power levels of the transmitted and reflected beams detected by these photodetectors **4202**, **4204** can be compared in the electronic domain by circuitry and/or a processor (not shown) to provide an indication of the range from the LiDAR to the scene.

[0183] Some embodiments of the photonic chips may include tunable optical filters connected to the photodetectors to filter the reflected light (not shown in FIG. **42**). These tunable optical filters may be used to preferentially transmit light at the wavelengths of the NIR lasers and reject ambient light coupled into the photonic chip via the grating(s). The tunable optical filters may be implemented as cascaded add-drop ring resonator filters, photonic crystal filters, evanescent directional couplers, echelle gratings, arrayed waveguide gratings, or corrugated sidewall grating filters. Integrated resistive heaters positioned above or laterally offset from the filter devices may be used to thermo-optically tune the filters to align the transmission spectra of the filters to the NIR laser wavelengths. In other embodiments, the waveguides receiving the reflected light from the gratings via the slabs may be separate waveguides with small lateral offsets from the corresponding waveguides transmitting light into the slab, grating, and then free space. In this case, the tunable optical filters and photodetectors can be directly connected to each receiving waveguide.

[0184] FIGS. **43A** and **43B** show photonic integrated circuit chips **4300a** and **4300b**, respectively, used for LiDAR. Each photonic chip may include a reflector on the movable plate (e.g., thin film metal mirror over the slab region of the plate) and an external reflector, and in this approach, the optical deflection angle is three times the mechanical deflection of the movable plate.

[0185] FIG. 43A illustrates a photonic integrated circuit chip **4300a** being used for LiDAR, with NIR beams emitted by the grating(s) **150** on the movable plate **140** scanned across a scene by the movable plate **140**. At each position of the plate, reflections of the beams from objects in the scene are coupled back into the photonic chip **4300a** via their respective grating coupler(s) **150**.

Waveguides in the photonic chip **4300a** couple the reflected beams to a receiver photonic circuit **4302** (e.g., balanced photodetector) integrated with the photonic chip **4300a**.

[0186] FIG. 43B shows a LiDAR system with an off-chip reflector **4306** and a reflector **4304** integrated onto the movable plate **140** (e.g., a metal mirror above the waveguide slab region of the plate). The transmitted beams from the grating(s) **150** are reflected by the off-chip mirror **4306**, directed toward the reflector **4304** on the movable plate **140**, and reflected toward the scene. The movable plate **140** scans the beams, and the deflection of the optical beam is three times the deflection of the movable plate **140**. Reflected beams from the scene pass through the same paths back to the photonic chip **4300b** and are coupled into a receiver photonic circuit **4302** in the photonic chip **4300b** via their respective grating coupler(s) **150**.

[0187] Like the light-engine system shown in FIGS. 37 and 38, additional variations for LiDAR include photonic chips with slabs and gratings not on a movable plate. Instead, the beams are scanned by a two-axis MEMS mirror integrated onto the photonic chip or on a separate chip. The beams may also be scanned by two one-axis MEMS mirrors (each on a separate chip or one on a separate chip and one integrated onto the photonic chip).

[0188] FIG. 44 shows some of these LiDAR system variations. Here, the gratings **150** and slabs **152** on the photonic chips **4400a**, **4400b** are not integrated onto a movable plate. (Top left) One or more beams from the grating **150** are directed by an off-chip reflector **4402** toward a 2-axis MEMS mirror **4404** integrated onto the photonic chip **4400a**. The MEMS mirror **4404** scans the beams. (Bottom left) The 2-axis MEMS mirror **4414** is on a separate chip mounted on the same common substrate/carrier **202** with the photonic chip **4400b**. (Top right) Two separate 1-axis MEMS mirror chips **4406**, **4408** are used to scan the emitted beams from the photonic chip **4400b**. (Bottom right) One 2-axis MEMS mirror chip **4418** with an on-chip reflector **4420** is positioned above the photonic chip **4400b** and used to scan the beams. In all cases, reflected beams pass backward through the system and are coupled onto the photonic chip **4400a**, **4400b** via their respective grating coupler(s) **150**.

[0189] Other embodiments of LiDAR optical systems may include separate transmitter and receiver modules. The transmitter and receiver modules may have the same configurations as in FIGS. 43A, 43B, and 44. During operation the movable plates and/or scanning mirrors of the two modules may be synchronized so that as the transmitter scans beams across a scene, reflected light is coupled into corresponding waveguides on the receiver photonic integrated circuit.

[0190] The photonic integrated circuit chips for LiDAR systems can also include the variations using details described above with respect to FIGS. 1-36, e.g., laser, photodetector, thermal sensor, strain sensor, grating, slab, movable plate, actuator, and edge coupler variations, embodiments with optical switches and/or modulators, and variations with actuators on the movable plate to compensate for deposited thin film stress.

[0191] Systems with Simultaneous Light Engine and LiDAR Functionalities Augmented reality glasses can use both display and 3D imaging functionalities. FIG. 45A-45C show optical systems based on photonic integrated circuits with simultaneous light engine and LiDAR functionalities.

[0192] FIG. 45A shows a photonic chip **4500** for simultaneous light engine and LiDAR functionalities, with some circuit elements and the routing waveguides omitted for clarity. It includes directly modulated R, G, B, and NIR lasers **120**, **4120** flip-chip bonded onto or otherwise co-packaged with the photonic chip **4500**. Subsets of R, G, B, and NIR waveguides (not shown) couple light to one or more slabs **152** for in-plane beam expansion and one or more gratings **150** emit the beams out of the photonic chip **4500**. The grating(s) **150** and slab(s) **152** are integrated onto a movable plate **140**, which steers the RGB and NIR beams in 2D. In addition, reflected NIR

light is collected by the gratings **150** and detected by photonic circuitry on the chip. Tunable reflectors **4502**, frequency discriminators, and receiver circuits (e.g., balanced photodetectors **4302**) on the chip enable time-of-flight, AMCW, or FMCW LiDAR.

[0193] FIGS. **45B** and **45C** show different configurations of the photonic chip **4500** of FIG. **45A** positioned close to the input coupler **106** of a waveguide combiner **108**. Scanned RGB beams are coupled into the waveguide combiner **108**, while scanned NIR beams pass through the waveguide combiner **108** and are scanned across the scene. Reflected NIR beams pass through the waveguide combiner **108**, are coupled back onto the photonic chip **4500** via their respective grating(s) **150** and detected (either with individual or balanced on-chip photodetectors). Additional optical elements may be placed on the outside of the waveguide combiner and in the path of the NIR beams to correct for aberrations and/or distortions in the NIR beams or widen the scan angle (FOV) of the NIR beams (at the expense of increasing beam divergence) (not shown in FIGS. **45B** and **45C**).

[0194] FIG. **46** shows a photonic integrated circuit chip and system configuration with gratings **4602** and slabs (not shown) arranged along a curve on the photonic chip. The curve and emission angles of the gratings **4602** are selected such that the beams can be reflected by an off-chip reflector **4604** above the chip and directed toward a 2-axis MEMS mirror **4606** either integrated onto the photonic chip or on a separate chip (top-right). The MEMS mirror **4606** scans the beams and directs them toward the input coupler **4608** of an optical combiner. Alternatively, the external MEMS mirror **4606** may be positioned above the photonic chip (bottom-right). The beams may directly overlap at the MEMS mirror plane or directly overlap at a plane slightly above the MEMS mirror plane (close to or at the input coupler plane).

Configuration, Fabrication, and Substrate Details

[0195] The photonic chips described herein may be fabricated on a number of wafer substrates. The chips may be fabricated on bulk Si, silicon on insulator (SOI), or double SOI wafers. SOI wafers may further be divided into wafers with a thin Si device layer (thickness $<1\ \mu\text{m}$, often used for Si photonics applications) and wafers with a thick Si device layer (thickness about 5-150 μm , often used for MEMS applications). Trenches may be etched into the front of the wafer to define the shapes of the movable plate (or integrated MEMS mirror), torsion springs, and edge coupler facets. The whole wafer may be thinned (e.g., by back grinding or etching) to thin and separate the chips from the wafer (resulting in chips with a uniform thickness). Alternatively, the backside of the wafer may be patterned with trenches to selectively thin the torsion springs and movable plate (or integrated MEMS mirror).

[0196] As mentioned above, the overhead on movable plate size due to the slab waveguide leads to plate sizes larger than the emitted beam diameter. Dynamic deformation due to the plate size may be addressed using known methods for MEMS mirrors. For example, ridges may be patterned onto the back of the movable plate such that the movable plate is effectively thicker (and more resistant to dynamic deformation) without incurring the same increase in mass as if the movable plate had a large uniform thickness. In addition, curvature of the movable plate (due to stresses of the dielectric waveguide and cladding layers deposited on top) may be compensated by the stress of a dielectric layer on the back of the movable plate.

CONCLUSION

[0197] While various inventive embodiments have been described and illustrated herein, those of ordinary skill in the art will readily envision a variety of other means and/or structures for performing the function and/or obtaining the results and/or one or more of the advantages described herein, and each of such variations and/or modifications is deemed to be within the scope of the inventive embodiments described herein. More generally, those skilled in the art will readily appreciate that all parameters, dimensions, materials, and configurations described herein are meant to be exemplary and that the actual parameters, dimensions, materials, and/or configurations will depend upon the specific application or applications for which the inventive teachings is/are used. Those skilled in the art will recognize or be able to ascertain, using no more than routine

experimentation, many equivalents to the specific inventive embodiments described herein. It is, therefore, to be understood that the foregoing embodiments are presented by way of example only and that, within the scope of the appended claims and equivalents thereto, inventive embodiments may be practiced otherwise than as specifically described and claimed. Inventive embodiments of the present disclosure are directed to each individual feature, system, article, material, kit, and/or method described herein. In addition, any combination of two or more such features, systems, articles, materials, kits, and/or methods, if such features, systems, articles, materials, kits, and/or methods are not mutually inconsistent, is included within the inventive scope of the present disclosure.

[0198] Also, various inventive concepts may be embodied as one or more methods, of which an example has been provided. The acts performed as part of the method may be ordered in any suitable way. Accordingly, embodiments may be constructed in which acts are performed in an order different than illustrated, which may include performing some acts simultaneously, even though shown as sequential acts in illustrative embodiments.

[0199] All definitions, as defined and used herein, should be understood to control over dictionary definitions, definitions in documents incorporated by reference, and/or ordinary meanings of the defined terms.

[0200] The indefinite articles “a” and “an,” as used herein in the specification and in the claims, unless clearly indicated to the contrary, should be understood to mean “at least one.”

[0201] The phrase “and/or,” as used herein in the specification and in the claims, should be understood to mean “either or both” of the elements so conjoined, i.e., elements that are conjunctively present in some cases and disjunctively present in other cases. Multiple elements listed with “and/or” should be construed in the same fashion, i.e., “one or more” of the elements so conjoined. Other elements may optionally be present other than the elements specifically identified by the “and/or” clause, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, a reference to “A and/or B”, when used in conjunction with open-ended language such as “comprising” can refer, in one embodiment, to A only (optionally including elements other than B); in another embodiment, to B only (optionally including elements other than A); in yet another embodiment, to both A and B (optionally including other elements); etc.

[0202] As used herein in the specification and in the claims, “or” should be understood to have the same meaning as “and/or” as defined above. For example, when separating items in a list, “or” or “and/or” shall be interpreted as being inclusive, i.e., the inclusion of at least one, but also including more than one, of a number or list of elements, and, optionally, additional unlisted items. Only terms clearly indicated to the contrary, such as “only one of” or “exactly one of,” or, when used in the claims, “consisting of,” will refer to the inclusion of exactly one element of a number or list of elements. In general, the term “or” as used herein shall only be interpreted as indicating exclusive alternatives (i.e., “one or the other but not both”) when preceded by terms of exclusivity, such as “either,” “one of,” “only one of,” or “exactly one of.” “Consisting essentially of,” when used in the claims, shall have its ordinary meaning as used in the field of patent law.

[0203] As used herein in the specification and in the claims, the phrase “at least one,” in reference to a list of one or more elements, should be understood to mean at least one element selected from any one or more of the elements in the list of elements, but not necessarily including at least one of each and every element specifically listed within the list of elements and not excluding any combinations of elements in the list of elements. This definition also allows that elements may optionally be present other than the elements specifically identified within the list of elements to which the phrase “at least one” refers, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, “at least one of A and B” (or, equivalently, “at least one of A or B,” or, equivalently “at least one of A and/or B”) can refer, in one embodiment, to at least one, optionally including more than one, A, with no B present (and optionally including elements other than B); in another embodiment, to at least one, optionally including more than one, B, with

no A present (and optionally including elements other than A); in yet another embodiment, to at least one, optionally including more than one, A, and at least one, optionally including more than one, B (and optionally including other elements); etc.

[0204] In the claims, as well as in the specification above, all transitional phrases such as “comprising,” “including,” “carrying,” “having,” “containing,” “involving,” “holding,” “composed of,” and the like are to be understood to be open-ended, i.e., to mean including but not limited to. Only the transitional phrases “consisting of” and “consisting essentially of” shall be closed or semi-closed transitional phrases, respectively, as set forth in the United States Patent Office Manual of Patent Examining Procedures, Section 2111.03.

Claims

1. A photonic integrated circuit chip comprising: a substrate; a movable plate suspended from the substrate and configured to tilt about a first axis with respect to the substrate and about a second axis with respect to the substrate; an array of lasers, integrated with the substrate, to emit red light, blue light, and green light; an array of waveguides, integrated with the substrate in optical communication with the array of lasers, to guide the red light, the blue light, and the green light to the movable plate; a slab waveguide, integrated with the movable plate, to guide the red light, the green light, and the blue light from the array of waveguides; a grating, integrated with the movable plate in optical communication with the slab waveguide, to emit the red light, the green light, and the blue light into free space; and at least one actuator, operably coupled to the movable plate, to tilt the movable plate about the first axis and about the second axis so as to scan the red light, the green light, and the blue light across at least a portion of a field of view.
2. The photonic integrated circuit chip of claim 1, wherein the substrate forms an outer frame around the movable plate, and further comprising: an inner frame suspended from the substrate and configured to tilt about the first axis with respect to the substrate, wherein the movable plate is suspended from the inner frame and configured to tilt about the second axis with respect to the substrate and the inner frame.
3. The photonic integrated circuit chip of claim 2, further comprising: a conductive coil, integrated with the inner frame, to cause two-dimensional rotation of the movable plate via the Lorentz force in response to a current.
4. The photonic integrated circuit chip of claim 2, further comprising: inner torsion springs mechanically coupling the movable plate to the inner frame; and outer torsion springs mechanically coupling the inner frame to the outer frame.
5. The photonic integrated circuit chip of claim 1, wherein the slab waveguide is configured to form the red light in multiple beams of red light, the green light into multiple beams of green light, and the blue light into multiple beams of blue light.
6. The photonic integrated circuit chip of claim 5, wherein the grating is configured to emit the multiple beams of red light, the multiple beams of green light, and the multiple beams of blue light into free space and the at least one actuator and the movable plate are configured to scan the multiple beams of red light, the multiple beams of green light, and the multiple beams of blue light across different slices of the field of view.
7. The photonic integrated circuit chip of claim 1, wherein the grating is an apodized grating.
8. The photonic integrated circuit chip of claim 1, wherein the grating is curved in a plane of the movable plate.
9. The photonic integrated circuit chip of claim 1, wherein the grating is configured to emit the red light, the green light, and the blue light as collimated beams.
10. The photonic integrated circuit chip of claim 1, wherein the grating comprises: cladding material; a first layer of grating teeth, embedded in the cladding material, to diffract first portions of the red light, the blue light, and the green light into free space at first angles with respect to the

movable plate; and a second layer of grating teeth, embedded in the cladding material, to diffract second portions of the red light, the blue light, and the green light into free space at second angles with respect to the movable plate.

11. The photonic integrated circuit chip of claim 1, wherein the slab waveguide is a first slab waveguide configured to guide first portions of the red light, the green light, and the blue light, the grating is a first grating configured to emit first portions of the red light, the green light, and the blue light into free space in first directions, and further comprising: a second slab waveguide, integrated with the movable plate in optical communication with the array of waveguides, to guide second portions of the red light, the green light, and the blue light; and a second grating, integrated with the movable plate in optical communication with the second slab waveguide, to emit the second portions of the red light, the green light, and the blue light into free space in second directions.

12. The photonic integrated circuit chip of claim 11, wherein the first slab waveguide overlaps the second slab waveguide.

13. The photonic integrated circuit chip of claim 12, wherein the first grating overlaps the second grating.

14. The photonic integrated circuit chip of claim 1, wherein the at least one actuator comprises: cantilevered piezoelectric actuators coupling the movable plate to the substrate and configured to rotate the movable plate about the first axis and the second axis.

15. The photonic integrated circuit chip of claim 1, wherein the array of lasers is further configured to emit infrared light, the array of waveguides is configured to guide the infrared light to the movable plate, the slab waveguide is configured to guide the infrared light to the grating, and the grating is configured to emit the infrared light into free space, and further comprising: at least one photodetector, in optical communication with the grating, to detect returned infrared light from an object illuminated by the infrared light.

16. A light engine for an augmented reality display, the light engine comprising: the photonic integrated circuit chip of claim 1; and an optical combiner, in optical communication with the photonic integrated circuit chip, to direct the red light, the green light, and the blue light emitted into free space by the grating toward an eye of a person viewing the augmented reality display.

17. The light engine of claim 16, wherein the array of lasers is further configured to emit infrared light, the array of waveguides is configured to guide the infrared light to the movable plate, the slab waveguide is configured to guide the infrared light to the grating, and the grating is configured to emit the infrared light into free space, and the optical combiner is configured to transmit the infrared light onto a scene, and further comprising: at least one photodetector to detect returned infrared light from the scene.

18. A light engine for an augmented reality display, the light engine comprising: the photonic integrated circuit chip of claim 1; and a mirror, in optical communication with the grating, to reflect the red light, the green light, and the blue light.

19. A photonic integrated circuit chip comprising: a substrate; a movable plate suspended from the substrate and configured to tilt about a first axis with respect to the substrate and about a second axis with respect to the substrate; an array of lasers, integrated with the substrate, to emit infrared light; an array of waveguides, integrated with the substrate in optical communication with the array of lasers, to guide the infrared light to the movable plate; a slab waveguide, integrated with the movable plate, to guide the infrared light from the array of waveguides; a grating, integrated with the movable plate in optical communication with the slab waveguide, to emit the infrared light into free space; at least one actuator, operably coupled to the movable plate, to tilt the movable plate about the first axis and about the second axis so as to scan the infrared light across at least a portion of a field of view; and at least one photodetector, in optical communication with the grating, to detect returned infrared light from an object illuminated by the infrared light.

20. A LiDAR system comprising: the photonic integrated circuit chip of claim **19**; and a mirror, in optical communication with the grating, to reflect the infrared light.
